



Project 5:

Credit Value-at-Risk

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Credit Risk

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1. Compute the Credit VaR again, now assuming that the correlation between the two bonds is 0.072.

Year-end bond price vectors

For each destination rating k , the year-end value of bond j is the present value of its remaining cash flows discounted at the one-year forward zero rates for rating k from Table 4.2 of the handouts:

$$P_{j(k)} = \sum_{t=1}^{M_j} CF_{j,t} \cdot D(k, t)$$

$$D(k, t) = \frac{1}{(1 + r_{k,t})^t}$$

Here $r_{k,t}$ is the one-year forward zero rate for rating k at horizon t and M_j is the maturity of bond j . For bonds assigned to the Default state the value equals the recovery amount $R = 51.13$.

Rating-change thresholds

The standardised asset return r of each obligor is modelled as $N(0,1)$. For a bond with initial rating, we define the threshold z_k as the z-score such that realising $r > z_k$ is consistent with migrating to rating k or better. Formally:

$$z_k = \Phi^{-1} \left(\sum_{j \geq k} p_j \right)$$

where p_j is the one-year transition probability to rating j and the sum accumulates all states j worse than k . Equivalently, $z_k = \Phi^{-1}(1 - \sum_{j \leq k} p_j)$. The resulting thresholds for the two bonds are given in Table 2, where z_k is the lower boundary of rating class k .

Joint transition probability matrix

The standardised returns of the two bonds are jointly bivariate normal with correlation $\rho = 0.072$. The probability that Bond 1 migrates to rating i and Bond 2 to rating k simultaneously is:

$$p_{ik} = \Phi_2(z_i, z_k; \rho) - \Phi_2(z_{i+1}, z_k; \rho) - \Phi_2(z_i, z_{k+1}; \rho) + \Phi_2(z_{i+1}, z_{k+1}; \rho)$$

where $\Phi_2(\cdot, \cdot; \rho)$ is the bivariate standard normal CDF with correlation ρ , and z_{k+1} is defined as $-\infty$ for the Default state. This yields an 8×8 joint probability matrix.

Expected portfolio value and Credit VaR

The expected portfolio value uses the marginal transition probabilities of each bond, exploiting linearity of expectation:

$$\mu_V = \sum_i p_i^{(1)} P_1(i) + \sum_k p_k^{(2)} P_2(k)$$

The full joint distribution of portfolio values $V = P_1(i) + P_2(k)$ is built from the 8×8 joint probability matrix. Values are sorted in ascending order and the 1% quantile V^* is the largest value whose cumulative probability does not exceed 0.01. The Credit VaR is then:

$$\text{Credit VaR} = \mu_V - V^*$$

Results

Applying the procedure with $\rho = 0.072$ yields:

$$\mu_V = 213.27$$

$$V^* = 203.73$$

$$\text{Credit VaR} = 9.54$$

The expected portfolio value $\mu_V = 213.27$ is unchanged relative to the warm-up ($\rho = 0.30$) because marginal transition probabilities are independent of correlation. The 1% quantile $V^* = 203.73$ and $\text{Credit VaR} = 9.54$ coincide with the warm-up result: the worst scenario on the discrete 8×8 grid is driven by joint downgrades that have essentially the same probability at both $\rho = 0.072$ and $\rho = 0.30$.

2. **Compute the VaR through simulations. Assume that Bond 1 (rated BBB) is issued by Vivendi and Bond 2 (rated A) is issued by GM. Use the same data as in example 4.1.2 of the handouts. Namely, standardized returns are given by:**

$$r_{Viv} = 0.4T + 0.2L + \epsilon_{Viv}$$

$$r_{GM} = 0.9C + \epsilon_{GM}$$

Assume for each company i that $\epsilon_i \sim iid N(0, \sigma_{\epsilon_i}^2)$. Proceed in the following steps:

1. Compute the variance of the specific risks, $\sigma_{\epsilon_{Viv}}$ and $\sigma_{\epsilon_{GM}}$.

2. Generate 100,000 scenarios and compute the Credit VaR.

We compute the Credit VaR via Monte Carlo simulation for the same two-bond portfolio, now modelling asset returns through the factor structure of Example 4.1.2. The standardised returns are:

$$r_{Viv} = 0.4T + 0.2L + \epsilon_{Viv}$$

$$r_{GM} = 0.9C + \epsilon_{GM}$$

The three common factors (T, L, C) are jointly $N(0, \Sigma_F)$ with the correlation matrix given in the handouts ($\rho_{TL} = 0.30, \rho_{TC} = 0.10, \rho_{LC} = 0.20$). The idiosyncratic shocks ϵ_i are drawn independently from $N(0, \sigma_{\epsilon_i}^2)$.

Step 1 - Specific risk variances

Because all returns are standardised to unit variance, the total variance of r_i decomposes as:

$$\text{Var}(r_i) = b_i^T \Sigma_F b_i + \sigma_{\epsilon_i}^2 = 1$$

Solving for the idiosyncratic variance:

$$\sigma_{\epsilon_i}^2 = 1 - b_i^T \Sigma_F b_i$$

For Vivendi (loadings $\beta_T = 0.4, \beta_L = 0.2, \beta_C = 0$):

$$b_{Viv}^T \Sigma_F b_{Viv} = 0.4^2 + 0.2^2 + 2 \cdot 0.4 \cdot 0.2 \cdot \rho_{TL} = 0.16 + 0.04 + 0.048 = 0.2480$$

$$\sigma_{Viv} = \sqrt{1 - 0.2480} = \sqrt{0.7520} = 0.8672$$

For General Motors (loadings $\beta_T = 0, \beta_L = 0, \beta_C = 0.9$):

$$b_{GM}^T \Sigma_F b_{GM} = 0.9^2 \cdot \text{Var}(C) = 0.81$$

$$\sigma_{GM} = \sqrt{1 - 0.81} = \sqrt{0.19} = 0.4359$$

Step 2 - Simulation ($N = 100,000$ scenarios)

In each scenario s , three correlated factor draws $(T_s, L_s, C_s) \sim MVN(0, \Sigma_F)$ and two independent idiosyncratic shocks $\epsilon_{Viv,s} \sim iid N(0, \sigma_{Viv}^2)$, $\epsilon_{GM,s} \sim iid N(0, \sigma_{GM}^2)$ are generated. The standardised returns are computed from the factor equations, and each return is mapped to a year-end rating by comparison with the z-score thresholds from Step 2 of Exercise 1. The portfolio value in scenario s is:

$$V_s = P_1(k_{1,s}) + P_2(k_{2,s})$$

where $k_{j,s}$ is the year-end rating assigned to bond j in scenario s . Sorting the N portfolio values in ascending order, V^* is the value at rank $\lfloor N \times 0.01 \rfloor = 1,000$.

Results

The simulation with $N = 100,000$ scenarios and random seed 42 gives:

$$\sigma_{Viv} = 0.8672$$

$$\sigma_{GM} = 0.4359$$

$$\mu_V = 213.27$$

$$V^* = 204.39$$

$$\text{Credit VaR} = 8.88$$

The simulation yields an exact 99% VaR rather than a conservative grid-based estimate, which explains why $\text{Credit VaR} = 8.88$ is slightly below the analytical 9.54 of Exercise 1.

- 3. Compute the Credit VaR through simulations for the portfolio with all the bonds listed in the xls file. Assume that the Recovery rates are constant (for each bond, set R to the R mean value given in the data file).**

We extend the simulation to the full portfolio of 17 bonds listed in the data file, each characterised by its rating, maturity, coupon rate, recovery rate, and loadings on the three common factors. Recovery rates are held fixed at their mean values.

Factor model

The standardised asset return of bond j is driven by the same three common factors and a firm-specific shock:

$$r_j = \beta_{j,T}T + \beta_{j,L}L + \beta_{j,C}C + \varepsilon_j$$

The factor loadings $\beta_{j,T}, \beta_{j,L}, \beta_{j,C}$ are read from the data file for each bond. The idiosyncratic variance is:

$$\sigma_{\varepsilon,j}^2 = 1 - b_j^T \Sigma_F b_j$$

Simulation

In each of $N = 100,000$ scenarios, a single draw of the three correlated factors and 17 independent idiosyncratic shocks is generated. For each bond j , the return $r_{j,s}$ is mapped to a year-end rating via the bond-specific threshold vector (derived from the bond's initial rating and the transition matrix). The bond is then valued using the price vector constructed from Table 4.2 forward rates and the constant recovery rate R_j . The portfolio value in scenario s is:

$$V_s = \sum_{j=1}^{17} P_j(k_{j,s})$$

Results

With constant recovery rates and $N = 100,000$ scenarios:

$$\mu_V = 1,744.71$$

$$V^* = 1,485.77$$

$$\text{Credit VaR} = 258.94$$

The Credit VaR of 258.94 reflects the aggregate credit risk of a diversified 17-bond portfolio with total face value \$1,700. This represents approximately 15% of the expected portfolio value, driven primarily by the bonds with lower credit ratings (B, CCC) which carry the highest default and downgrade probabilities.

4. **Compute the VaR again through simulations for the portfolio with all the bonds, but now assume that the Recovery rate for each bond is random. In particular, assume a beta distribution with mean and standard deviation given in the data file. Hence, if a bond is in default in a given scenario, you will have to generate a Recovery value for that scenario.**

We repeat the simulation of Exercise 3 but replace the fixed recovery rate with a random draw from a Beta distribution whenever a bond defaults, capturing the empirical variation in recovery rates across default events.

Beta distribution for recovery rates

For bond j , the recovery rate R_j is modelled as $Beta(\alpha_j, \beta_j)$ when the bond defaults. The parameters are matched to the mean $m_j = R_{mean}/100$ and standard deviation $s_j = R_{std}/100$ provided in the data file via the method of moments:

$$c_j = \frac{m_j(1 - m_j)}{s_j^2} - 1$$

$$\alpha_j = m_j \cdot c_j$$

$$\beta_j = (1 - m_j) \cdot c_j$$

In each defaulted scenario, $R_{j,s} \sim Beta(\alpha_j, \beta_j)$ is drawn independently and the bond value for that scenario is set to $R_{j,s} \times 100$. Non-defaulting bonds retain the same pricing as in Exercise 3.

Results

With stochastic recovery rates and $N = 100,000$ scenarios:

$$\mu_V = 1,744.80$$

$$V^* = 1,459.52$$

$$\text{Credit VaR} = 285.28$$

Introducing stochastic recovery increases the Credit VaR from 258.94 (Exercise 3) to 285.28, a difference of 26.34. The expected portfolio value μ_V is essentially unchanged (1,744.80 vs 1,744.71), confirming that the Beta distribution is correctly centred on the mean recovery rates. The additional VaR arises because in adverse scenarios multiple bonds default simultaneously and draw low recovery realisations, amplifying tail losses beyond what fixed recoveries imply.